

MULTIPLAYER TURN-BASED GAME ENGINE USING JAVA

NAME: SUBHAM PAL

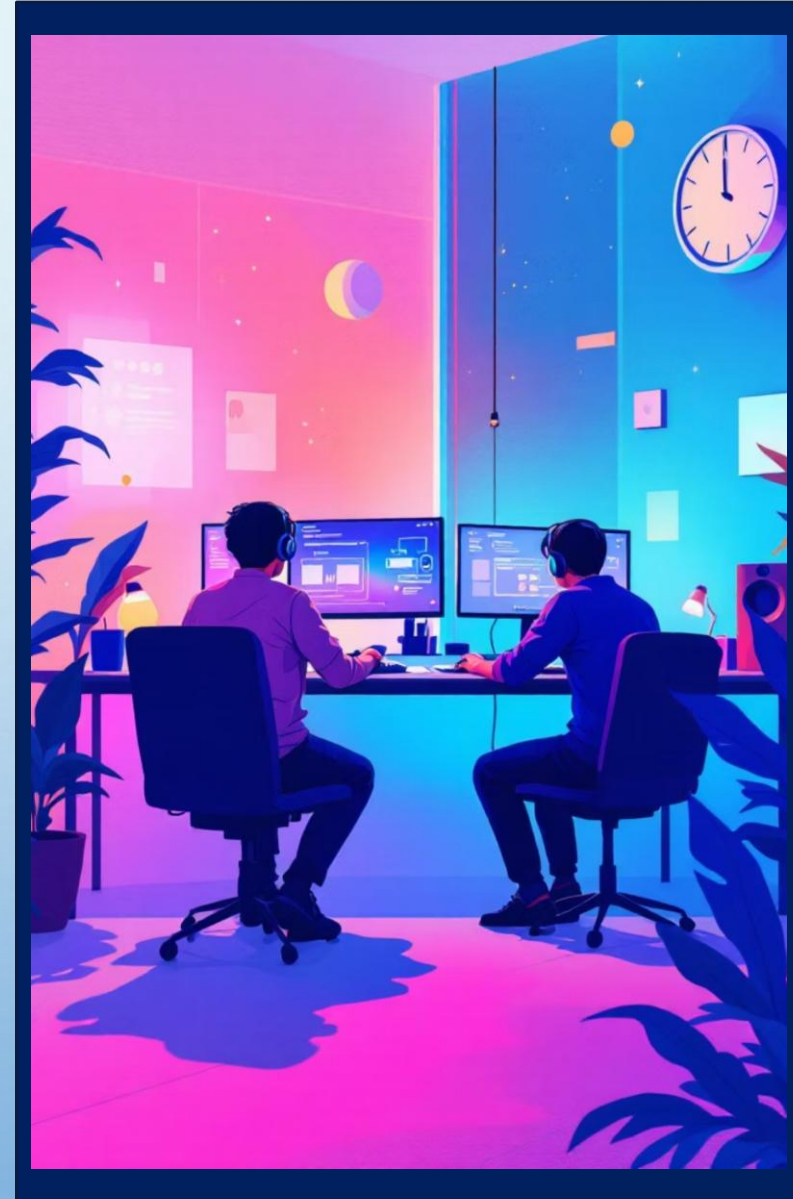
ROLL NO.: 82

PROBLEM:

LACK OF REUSABLE ARCHITECTURE IN TURN-BASED GAMES

TARGET USER:

STUDENTS AND BEGINNER DEVELOPERS



System Design

ARCHITECTURE:

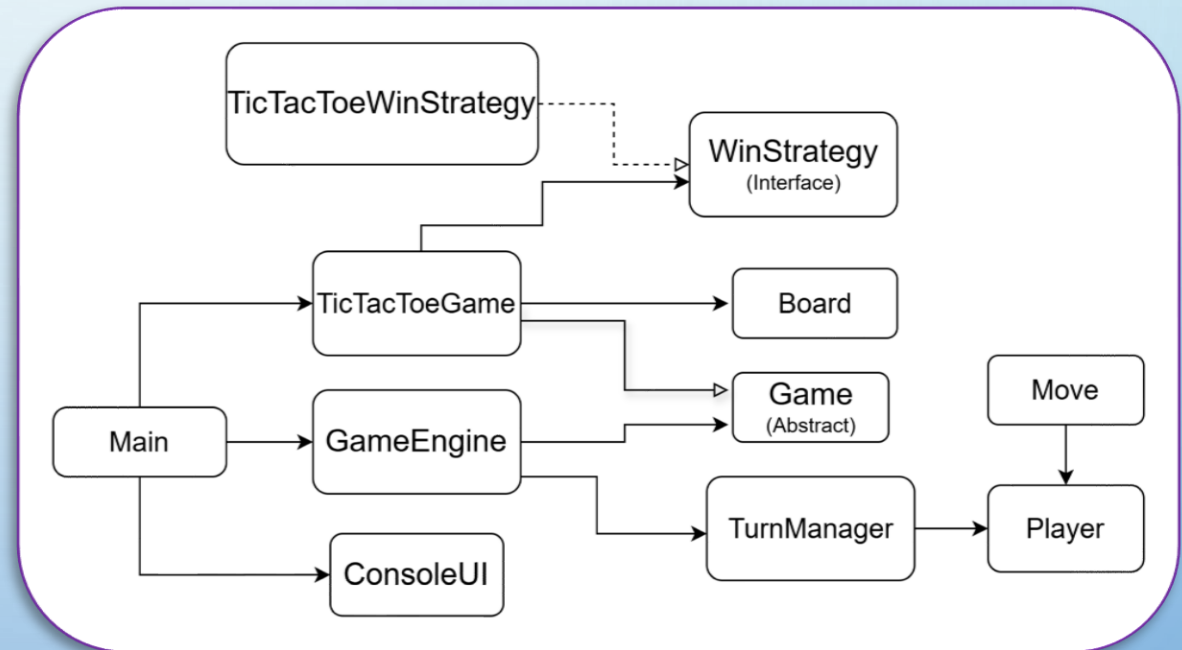
UI → ENGINE → GAME → STRATEGY → MODEL

KEY CLASSES:

- GAME
- TICTACTOEGAME
- GAMEENGINE
- TURNMANAGER
- PLAYER

PACKAGES:

APP, ENGINE, MODEL, GAMES, STRATEGY, UI



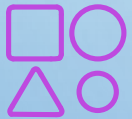
OOP Concepts Used



ABSTRACTION → GAME ABSTRACT CLASS



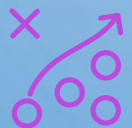
INHERITANCE → TICTACTOEGAME EXTENDS GAME



POLYMORPHISM → OVERRIDDEN METHODS (CHECKWINNER)



ENCAPSULATION → PLAYER CLASS (PRIVATE DATA)



DESIGN PATTERN → STRATEGY PATTERN (WINNER LOGIC)

TECHNICAL HIGHLIGHT

STRATEGY PATTERN



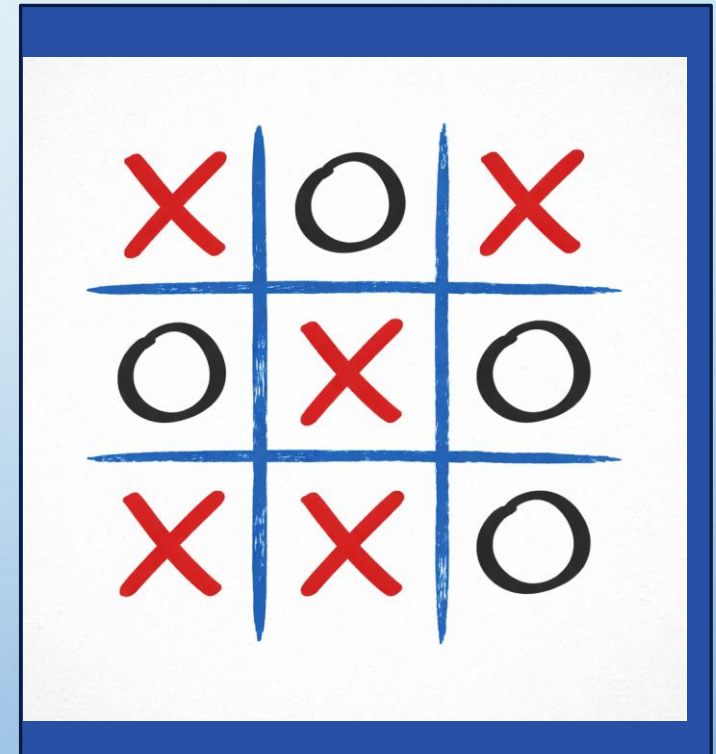
USED FOR WINNER DETECTION LOGIC



SEPARATES GAME RULES FROM ENGINE



ALLOWS EASY ADDITION OF NEW GAMES





Live Demo

- **GAME EXECUTION**
 - **VALID MOVE**
 - **INVALID MOVE (ERROR HANDLING)**
 - **WINNER DETECTION**
 - **DRAW CONDITION**
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